

CHOIBALSAN, Mongolia

WMO: 442590

Lat: 48.0811N

Lon: 114.5375E

Elev: 747

StdP: 92.67

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.8	-28.7	-34.7	0.2	-30.4	-32.6	0.2	-28.1	9.9	-16.2	8.9	-17.6	2.9	270	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	33.2	17.3	31.0	16.7	28.9	16.3	19.7	27.5	18.8	25.8	18.0	24.7	4.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	13.8	21.7	16.7	13.0	21.2	15.7	12.2	20.7	60.1	27.6	57.0	25.7	54.2	24.7	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.3	9.2	8.0	-32.5	36.4	2.9	2.7	-34.5	38.4	-36.2	40.0	-37.8	41.5	-39.9	43.5
			-32.8	21.2	3.0	1.0	-34.9	22.0	-36.6	22.6	-38.3	23.2	-40.4	23.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-20.2	-15.4	-6.4	4.5	12.2	18.9	21.7	19.4	12.6	2.8	-9.3	-17.5
	DBStd	15.50	4.96	5.57	6.50	5.16	4.77	4.18	3.23	3.58	4.70	5.31	6.65	5.06
	HDD10.0	4049	935	712	509	177	31	1	0	0	25	229	579	852
	HDD18.3	6196	1193	945	767	416	197	42	6	30	179	482	829	1111
	CDD10.0	1143	0	0	0	12	99	267	363	291	104	6	0	0
	CDD18.3	247	0	0	0	0	8	59	111	62	8	0	0	0
	CDH23.3	2711	0	0	0	12	158	721	1073	633	114	1	0	0
	CDH26.7	1023	0	0	0	2	51	282	427	233	29	0	0	0
Wind	WSAvg	3.5	3.4	3.4	3.7	4.4	4.4	3.4	3.0	2.9	3.3	3.4	3.3	3.4
Precipitation	PrecAvg	100	12	9	8	3	5	8	10	22	13	5	7	5
	PrecMax	289	106	100	52	23	31	67	102	143	107	37	63	39
	PrecMin	10	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	63	23	17	12	5	7	17	20	30	21	8	10	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.2	2.8	13.8	25.1	31.7	36.0	36.9	34.5	30.1	21.5	9.1	-2.6
		MCWB	-7.0	-1.4	4.3	9.8	13.5	16.7	19.4	18.2	15.1	10.0	2.7	-5.3
	2%	DB	-8.4	-1.6	9.8	20.1	27.5	32.5	33.4	31.6	26.7	16.9	5.6	-5.7
		MCWB	-9.8	-4.3	2.3	8.0	12.1	16.2	18.1	17.0	13.7	7.5	0.8	-7.6
	5%	DB	-10.7	-4.4	6.2	16.6	24.4	29.8	31.0	29.2	23.8	13.7	2.6	-8.1
		MCWB	-11.9	-6.4	0.4	6.1	11.3	15.6	17.9	16.4	12.9	5.8	-1.2	-9.5
	10%	DB	-12.9	-7.0	3.2	13.9	21.8	27.4	28.7	26.8	21.1	11.0	0.1	-10.3
		MCWB	-13.8	-8.5	-1.3	4.9	10.1	15.0	17.4	16.1	11.9	4.5	-2.8	-11.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.9	-1.4	5.1	10.1	15.0	18.7	21.4	20.4	17.4	10.3	3.1	-4.8
		MCDB	-5.4	2.7	13.0	22.9	28.4	28.1	30.1	29.4	26.0	20.4	8.6	-2.9
	2%	WB	-9.6	-4.0	2.6	8.2	13.3	17.7	20.3	19.3	15.5	8.1	1.0	-7.5
		MCDB	-8.3	-1.5	9.4	19.2	24.6	27.0	28.4	26.0	21.8	15.5	5.6	-5.7
	5%	WB	-11.7	-6.3	0.6	6.5	12.0	16.9	19.4	18.4	14.2	6.5	-0.9	-9.4
		MCDB	-10.6	-4.3	5.9	15.9	22.8	26.2	27.0	24.4	21.4	13.0	2.6	-8.1
	10%	WB	-13.7	-8.3	-1.1	5.0	10.7	16.1	18.6	17.5	12.8	5.0	-2.5	-11.3
		MCDB	-12.7	-6.7	3.0	13.5	20.6	24.7	25.6	23.6	20.0	10.7	0.2	-10.3
Mean Daily Temperature Range	5% DB	MDBR	8.9	11.0	11.4	12.5	13.5	12.2	11.0	11.6	12.4	11.2	10.0	8.5
		MCDBR	11.2	13.5	14.7	17.2	17.7	15.9	14.6	15.5	16.1	15.2	12.6	10.4
		MCWBR	10.0	11.1	9.2	9.0	7.9	5.9	4.7	5.4	7.3	8.7	8.8	8.8
	5% WB	MCDBR	11.0	13.3	14.2	16.4	16.0	12.9	11.7	11.2	13.4	14.0	12.3	10.1
		MCWBR	9.9	11.1	9.2	8.9	7.7	5.1	4.7	4.9	6.6	8.5	8.8	8.7
Clear-Sky Solar Irradiance	taub	0.256	0.278	0.319	0.387	0.456	0.458	0.422	0.382	0.337	0.308	0.275	0.260	
	taud	2.291	2.259	2.236	2.106	1.949	2.063	2.241	2.325	2.386	2.384	2.343	2.261	
	Ebn at Noon	826	882	895	859	808	809	836	857	868	837	793	762	
	Edh at Noon	72	94	115	146	178	160	132	116	99	83	68	66	
All-Sky Solar Radiation	RadAvg	1.91	3.13	4.55	5.64	6.41	6.48	6.12	5.62	4.55	3.13	1.98	1.45	
	RadStd	0.14	0.17	0.21	0.28	0.31	0.43	0.32	0.32	0.24	0.17	0.10	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	-2.20	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page